Labour Demand 1: Competitive Labour Markets

EC306- Wages and Employment

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Goals of This Section

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- Understand how firms decide how much labour to employ to produce
- Compare short-run vs. long-run labour demand decisions
- Explain why labour demand is downward sloping
- Learn factors that affect the elasticity of labour demand
- Understand the impact of free trade and international competition on the Canadian labour market

Background

Background

- Labour demand refers to the number of workers that firms want to hire at a given wage rate
- It is a **derived demand**
  - Linked to demand for the goods/services produced
  - It is an input into that production
- Several factors will influence labour demand
  - Firm's objectives and constraints
  - Demand conditions in product markets
  - State of technology
  - Time horizon
    - **Short run:** one or more inputs fixed (usually capital)
    - **Long run:** all inputs variable

Structure of Product Markets

- We noted that labour demand is driven by the firm's output
  - Labour is used as a input
- The structure of the product market will influence the demand for labour
  - Ranges from perfect competition (many sellers) to monopoly (one seller)
- By contrast, the structure of the labour market will influence the supply the firm faces
  - Ranges from perfect competition (many workers) to monopsony (one buyer)
- In this section we examine the case of perfect competition in both product and labour markets

Demand in the Short Run

Model

- As labour demand is derived from production, the model starts with a production function
  - Relates inputs to outputs

$$Q = f(K, N)$$

- Where
  - Q is quantity of output
  - K is capital (fixed in short run at K_0)
  - N is labour (variable in short run)
- Firms objective is to maximize profit
- In the short run this means choosing N given fixed K_0
- Optimality is where the value of the marginal product of labour equals its price

Model

- To see this mathematically, take output prices are taken as given (perfect competition).
- Profit

$$\pi(N) = pQ - wN - rK_0$$

$$= pF(K, N) - wN - rK_0$$

- Differentiate $\pi(N)$ w.r.t. N:

$$\frac{d\pi}{dN} = p\frac{\partial F}{\partial N}(K_0, N) - w$$

$$= pMP_N - w$$

Model

- Optimal N^* satisfies

$$pMP_N = w$$

- Where
  - MP_N is the marginal product of labour
  - pMP_N is the value of the marginal product of labour (VMP)
  - w is the wage rate (marginal cost of labour, MC_N)
- Optimality requires that the value of the last unit of labour hired equals its cost